Korea Technology and Information Promotion Agency for SMEs

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Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Hierarchical Factor Graph Optimization for Long-Horizon Multi-Robot Tasks			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Mechanical · Materials		Mechanical · Materials	
	Medium Category	Small Category	Medium Category	Small Category
	Robotics/Automation Machinery	Robot Control and Intelligence Technologies	*Robotics/Automation Machinery*	Robotics/Automation Machinery Software
Concept of Proposed Technology	The proposed technology aims to develop and validate the Hierarchical Factor Graph (HFG) framework to solve the planning-control gap in advanced mobile platforms. The system unifies hierarchical decomposition with probabilistic optimization, representing the robot's decision process as a pyramid of factor graphs, with each layer operating at a distinct temporal scale. This enables robust and computationally efficient long-horizon control for complex, multi-task settings on dynamic platforms.			

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	Key features: - A unified, multi-rate optimization framework expressed as a single nonlinear least-squares problem. - A learned autoencoder mechanism for differentiable abstraction between high-frequency, concrete states and low-frequency, abstract representations. - Scalable application from single-robot dynamic tasks (quadrupeds/humanoids) to coordinated multi-robot operations (teams of humanoids).

Rationale for Joint R&D Project	The successful translation of the HFG framework—which integrates cutting-edge probabilistic optimization with learned representation—into practical, high-performance robotic systems requires close collaboration. A joint R&D framework is essential to integrate these complex AI algorithms with real-world dynamic hardware.
	
	- Rationale for collaboration:
	  - Georgia Tech provides expertise in multi-rate HFG algorithms, learned abstraction, and large-scale optimization.
	  - The Korean SME partner provides capabilities in high-performance quadruped and humanoid hardware development, low-level system integration, and real-world deployment.
	  - Effective co-design is critical to meet the dynamic and computational demands of advanced platforms like humanoids and robot teams.
	- Expected benefits of joint R&D:
	  - Accelerated deployment of a computationally efficient and scalable planning architecture.
	  - Reduced technical risk by ensuring tight coupling between advanced control theory and industrial-grade hardware.
	  - Creation of a foundational technology for next-generation multi-robot intelligence and commercialization.

Key Words	1. Factor Graphs 2. Multi-Rate Planning and Control 3. Long-Horizon Multi-Task Robotics 4. Quadruped/Humanoid Locomotion 5. Inference via Optimization

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL) [2]	Start of the Project	Level (3)	End of the Project	Level (6)
R&D Objectives	The objective of this R&D project is to develop and validate the HFG framework, enabling highly efficient and reliable planning for long-horizon, complex tasks on advanced mobile platforms. The specific R&D objectives are: - To develop and validate the HFG framework for multi-rate planning and dynamic control for quadruped platforms. - To integrate learned state abstraction for long-horizon human-scale manipulation tasks on humanoid robots. - To extend the framework for scalable, distributed optimization to enable safe and coordinated operation for teams of humanoids.			
R&D Target Specifications	The target performance specifications of this R&D project are as follows: - **High-Frequency Control Stability (95%):** Achieve a successful dynamic task execution rate of greater than 95% for high-frequency control loops on physical quadruped and humanoid platforms in complex environments. - **Long-Horizon Multi-Step Success Rate (80%):** Achieve a success rate of at least 80% for 5+ step sequential manipulation and navigation tasks, demonstrating superior performance over monolithic planning methods. - **Planning Computational Efficiency (50%):** Achieve a computational time reduction of greater than 50% in the long-horizon planning cycle compared to monolithic control architectures, enabling high-rate, long-horizon decision-making.			

R&D Plan	This R&D project will be conducted over three years, following a progressive development and validation strategy from simulation to physical platforms and finally to multi-robot deployment.
	- **Year 1: HFG Core Development and Quadruped Platform Integration**
	  - Develop the core HFG framework and learned abstraction models.
	  - Implement a unified simulation environment for dynamic quadruped locomotion and long-horizon tasks.
	  - Validate core HFG algorithms through simulation-based experiments on quadruped tasks, focusing on dynamic stability and computational reduction.
	- **Year 2: Humanoid Adaptation and System Integration on Physical Quadruped**
	  - Integrate and customize the HFG framework onto the physical quadruped platform provided by the SME partner.
	  - Develop and test advanced learned abstractions for human-scale manipulation tasks on humanoids in simulation.
	  - Conduct repeated laboratory experiments to evaluate the robustness and multi-rate performance on the physical quadruped.
	- **Year 3: Real-World Deployment on Humanoid and Multi-Robot Team Scaling**
	  - Integrate the HFG system onto a physical humanoid platform and perform large-scale real-world task verification.
	  - Extend the framework to distributed HFG for teams of humanoids in simulation and initial real-world coordination experiments.
	  - Refine system reliability and assess commercialization potential for industrial transfer of multi-robot intelligence technology.			

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	Georgia Tech will be responsible for the development of the core HFG algorithms and the theoretical foundation for multi-rate, unified optimization. Georgia Tech will lead the research on learned abstraction and control methodologies required for efficient, long-horizon operation. The specific roles and responsibilities of Georgia Tech include: - Development of the HFG framework, including the pyramid of factor graphs and the learned autoencoder for state abstraction. - Design of the overall system architecture for multi-robot intelligence and the distributed optimization framework for robot teams. - Simulation-based development, algorithm validation, and performance analysis of the proposed technologies. - Technical guidance and collaboration to support system integration on physical platforms.
Expected R&R of Korean SME Partner	The Korean SME partner will be responsible for the development, integration, and validation of the dynamic robotic hardware and system-level implementation required. The SME will play a key role in translating the HFG algorithms into a practical and deployable system on high-performance platforms. The specific roles and responsibilities of the Korean SME partner include: - Design and development of quadruped and humanoid mobile manipulation hardware, including platforms, actuators, and sensor integration. - System integration of HFG algorithms with robotic hardware, embedded software platforms, and low-level control systems. - Execution of laboratory-scale and real-world experiments, including rigorous data collection, stress testing, and performance evaluation on the physical platforms. - Contribution to technology refinement, reliability improvement, and commercialization planning, leveraging industrial expertise and manufacturing capabilities.

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	The successful completion of this project will result in a fully validated, computationally-efficient, and scalable HFG framework capable of enabling advanced multi-rate planning for dynamic mobile platforms. This includes a prototype quadruped and humanoid robot running the HFG system, a comprehensive software package for learned state abstraction and unified optimization, and foundational results for multi-robot team coordination. The impact will be a significant leap in the capability of service and industrial robots to perform long-horizon, complex tasks with unprecedented stability and computational efficiency, positioning the Korean SME partner as a leader in next-generation robotic intelligence for humanoids and multi-robot systems.

6. References

1. Mukadam Mustafa, Dong Jing, Yan Xinyan, Dellaert Frank, Boots Byron. Continuous-time Gaussian process motion planning via probabilistic inference. The International Journal of Robotics Research. 2018; 37(11):1319--1340.

2. Mukadam Mustafa, Dong Jing, Dellaert Frank, Boots Byron. STEAP: simultaneous trajectory estimation and planning. Autonomous Robots. 2018; :1--20.

3. Dellaert Frank and Kaess Michael.. Factor graphs for robot perception. Foundations and Trends® in Robotics. 2017; 6(1-2):1--139.

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization

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